

ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Term Project Modeling Plan

Purpose

The modeling plan should explain how you intend to connect your occurrence and environmental data to your research question. Describe the model you will fit, the assumptions you will test, how you will evaluate the model, and how you will interpret its output.

This is a plan rather than a final methods section. You may revise it after fitting preliminary models. Your plan should distinguish relative environmental suitability from occupancy, abundance, or habitat quality.

Required content

1. Research question and intended interpretation

Restate your research question and identify the type of outcome you want the model to support. Explain whether your main interest is:

- Environmental associations
- Relative environmental suitability
- Predictor response shapes
- Geographic projection
- Transfer between regions or periods
- Change under past or future conditions

State what the model cannot establish from your available data.

2. Modeling data and sampling design

Briefly summarize the data entering the model:

- Final occurrence dataset and approximate sample size
- Definition of the calibration region or M
- Environmental predictors
- Number and placement of background locations
- Any correction for sampling bias or spatial clustering

Explain why the background represents environmental availability rather than confirmed absence.

3. Model choice

Identify your primary modeling method and explain why it fits your question and data. Examples include Maxent, a point-process model, GLM, GAM, boosted regression trees, or random forest. Identify at least one comparison model or alternative model configuration. The comparison may involve a different algorithm, level of complexity, predictor set, or assumption.

For each model, describe:

- The type of relationship it can represent
- Important assumptions
- Parameters that control model complexity
- Expected strengths and limitations

4. Model fitting and complexity

Explain how you will control model complexity and reduce overfitting. Depending on the method, this may include:

- Limiting predictor variables
- Choosing linear, quadratic, or smooth responses
- Adjusting regularization
- Restricting interactions
- Tuning tree depth or learning rate
- Comparing simpler and more flexible models

Explain how your occurrence sample size influences these decisions.

5. Model evaluation

Describe how you will divide or partition the data for model fitting and evaluation. Use spatially structured evaluation when feasible. If you plan to use random partitions, explain why.

Identify the evaluation measures you will use and what each one assesses. Possible measures include AUC, continuous Boyce index, omission rate, or comparison with withheld occurrences.

6. Robustness analysis

Identify at least two modeling decisions that you will vary. Possibilities include:

- Definition of M
- Occurrence filtering
- Background sampling
- Predictor selection
- Model algorithm
- Model complexity
- Data partitioning

State the biological interpretation you will compare across alternatives. Explain what result would cause you to describe that interpretation as stable, conditional, or not robust. When practical, change one major assumption at a time so that differences remain interpretable.

7. Prediction and interpretation

Describe the outputs you expect to produce, such as:

- Relative-suitability map
- Predictor response curves
- Variable-importance summary
- Model-evaluation table
- Comparison or uncertainty map

Explain how each output will help answer the research question. If projecting to another region or time period, describe how you will identify novel environments and limit unsupported extrapolation.

8. Reproducibility and feasibility

Identify the scripts, packages, and major processing steps needed to reproduce the analysis. State how you will record model settings, random seeds, package versions, and output files.

Identify one likely modeling problem and explain how you might address it. Include any questions on which you would like feedback.

Format and submission

- Due Friday, October 23 by 11:59 PM
- Up to three pages, single spaced
- PDF format
- 12-point Times New Roman
- Include citations for modeling methods or software
- Submit through the Modeling Plan assignment in Canvas